

NETWORKS AND COMPLEXITY

Solution 13-1

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 13.1: Quadratic polynomials [Lvl. 1]

Solve the following quadratic polynomials:

a) $1 + \lambda = \lambda^2$,

Solution

We solve this as

$$\lambda^2 - \lambda = 1 \quad (1)$$

$$\lambda^2 - \lambda + 1/4 - 1/4 = 1 \quad (2)$$

$$(\lambda - 1/2)^2 = 5/4 \quad (3)$$

$$\lambda - 1/2 = \pm\sqrt{5/4} \quad (4)$$

The result can be written as

$$\lambda = \frac{1 \pm \sqrt{5}}{2} \quad (5)$$

b) $(6 - \lambda)(2 - \lambda) - 60 = 0$,

Solution

Unfortunately we need to multiply this out

$$\lambda^2 - 8\lambda + 12 - 60 = 0 \quad (6)$$

$$(\lambda - 4)^2 + 12 - 60 - 16 = 0 \quad (7)$$

$$(\lambda - 4)^2 = 64 \quad (8)$$

$$\lambda - 4 = \pm 8 \quad (9)$$

Hence we find the two solutions $\lambda_1 = -4$ and $\lambda_2 = 12$

c) $(2 - \lambda)^2 - 9 = 0$.

Solution

This one is convenient as we already have a quadratic expression. So straight away

$$(2 - \lambda)^2 = 9 \quad (10)$$

$$\lambda - 2 = \pm 3 \quad (11)$$

and hence $\lambda_1 = 5$ and $\lambda_2 = -1$.